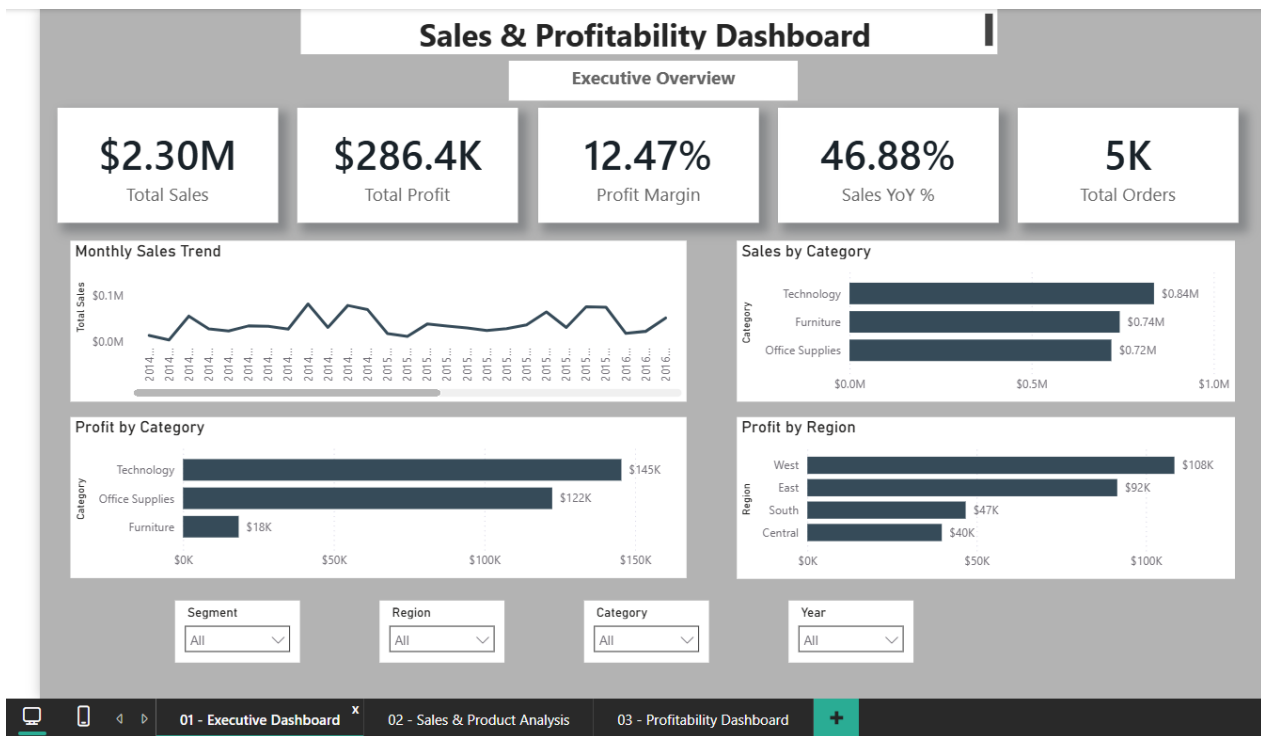


Superstore Sales & Profitability Analysis

End-to-End Business Intelligence Project

Microsoft Power BI • Power Query • DAX • Star Schema



Project Documentation

Prepared as a portfolio-ready Data Analytics project.

1. Project Overview

This project analyzes transactional sales data to understand sales performance, profitability, product and customer performance, geographic trends, and discount behavior. The solution was built as an end-to-end Power BI workflow, from data preparation and dimensional modeling through DAX, interactive dashboards, validation, and business recommendations.

Business Questions

- Where are sales and profit strongest?
- Which categories and products generate the most value?
- Which products or sub-categories are loss-making?
- How do customer segments and regions perform?
- Are higher discounts associated with weaker profitability?
- What actions could improve sustainable profitability?

2. Project Objectives

- Measure overall sales, profit, margin, orders, and year-over-year growth.
- Evaluate category, sub-category, product, customer, segment, region, and state performance.
- Identify high-sales/low-profit and loss-making areas.
- Build an interactive executive reporting experience for decision support.

3. Tools & Technologies

Tool / Technology	Purpose
Microsoft Power BI	Interactive dashboards and business visualization
Power Query	Data cleaning, transformation, and preparation
DAX	Measures, KPIs, and time-intelligence calculations
Data Modeling	Star schema, relationships, and surrogate keys
Excel / CSV	Source and supporting data

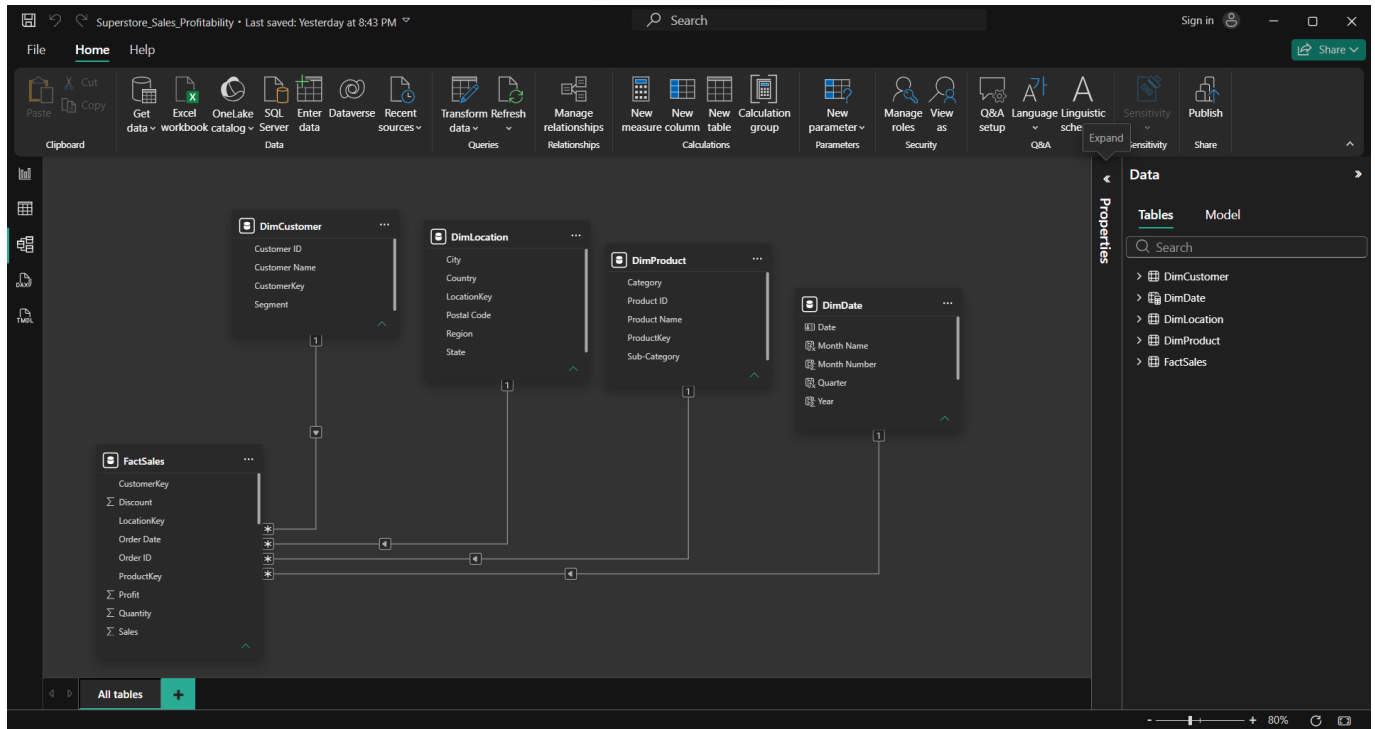
4. Data Preparation

The dataset was cleaned and prepared for analytical reporting, including data-type validation, field standardization, quality checks, and preparation for dimensional modeling.

5. Data Model

The model follows a Star Schema design. FactSales stores transactional measures, while dimension tables provide descriptive attributes for customers, products, locations, and dates. Surrogate keys were used to support reliable relationships between the fact and dimension tables.

6. Data Model — Final Model View



The final model separates transactional facts from descriptive dimensions and provides the foundation for consistent filtering, aggregation, and time-based analysis.

7. DAX & Key Measures

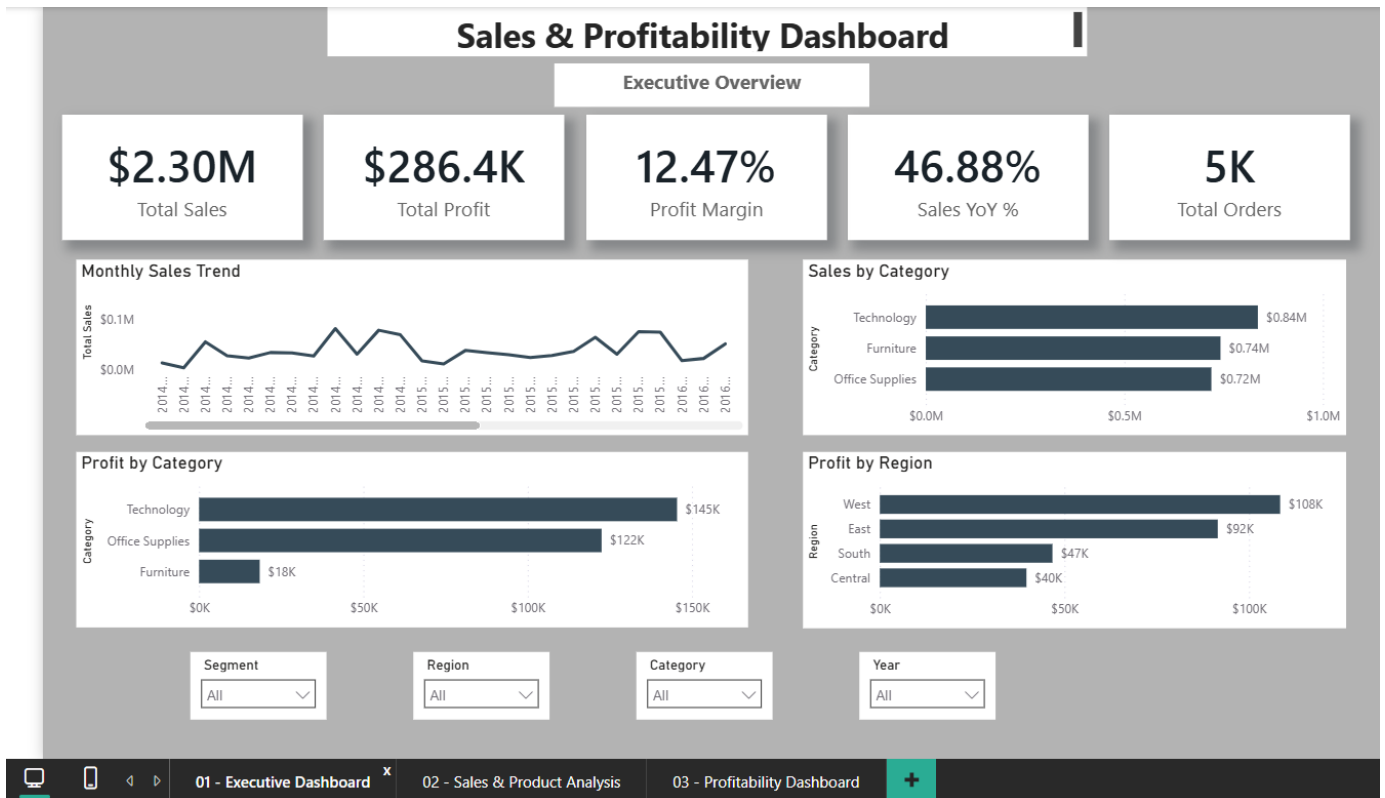
The report uses measures for core KPIs and time intelligence. Key measures include:

- Total Sales
- Total Profit
- Profit Margin
- Total Orders
- Total Customers
- Average Order Value
- Average Discount
- Sales YoY Growth
- YTD / time-intelligence calculations

8. Final Dashboard Structure

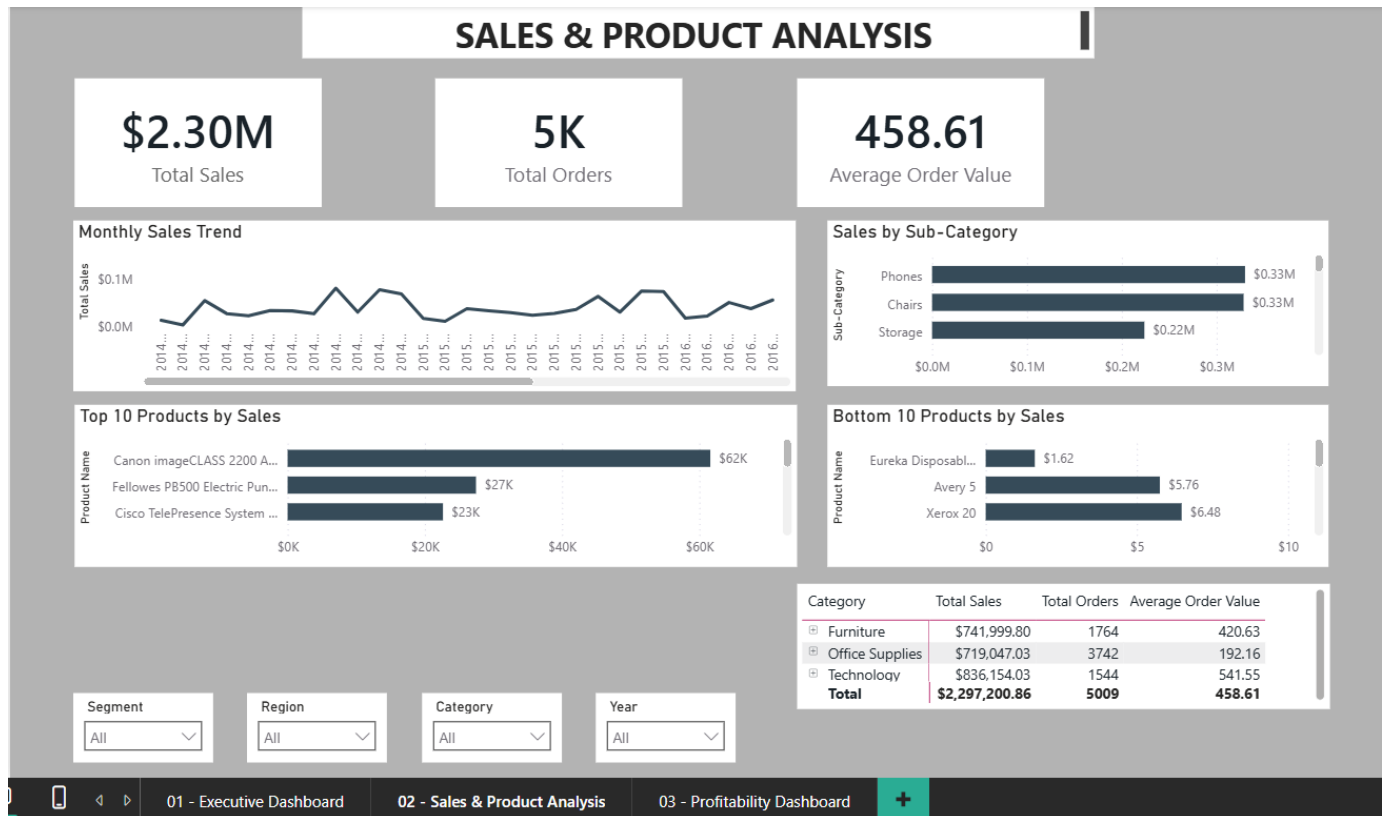
The final report was organized into three decision-oriented pages: an executive overview, a detailed sales and product analysis page, and a profitability analysis page.

9. Dashboard 01 — Executive Overview



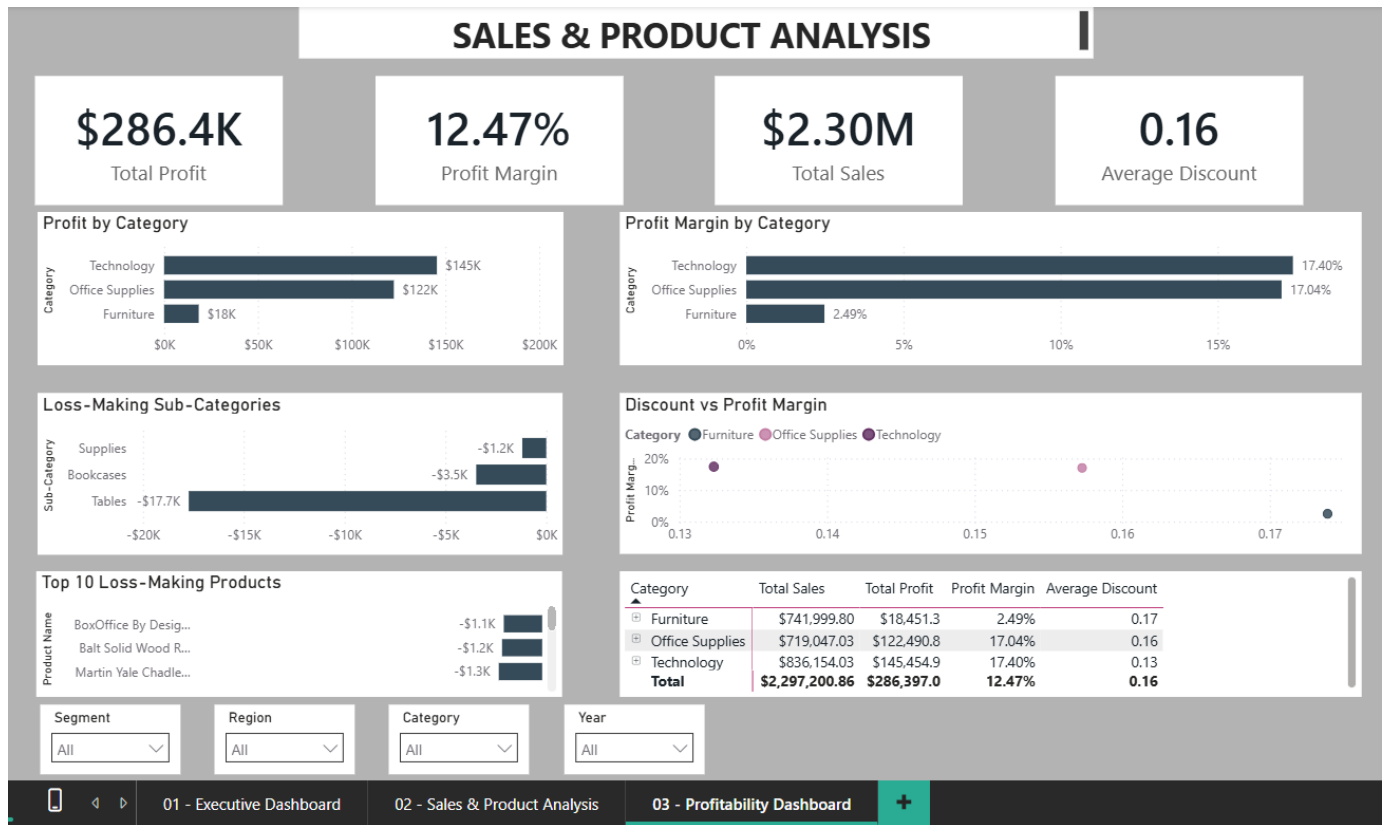
The executive page summarizes total sales, total profit, profit margin, YoY sales growth, orders, sales trend, category performance, and regional profitability.

10. Dashboard 02 — Sales & Product Analysis



This page focuses on sales trends, sub-category performance, top and bottom products by sales, and order-value metrics.

11. Dashboard 03 — Profitability Analysis



This page focuses on profit, profit margin, loss-making sub-categories, loss-making products, and the relationship between discount and profitability.

12. Key Business Insights

Overall performance: Total Sales are approximately \$2.30M, with Total Profit of approximately \$286.4K and an overall Profit Margin of 12.47%.

Growth: The dashboard reports Sales YoY Growth of 46.88%, indicating substantial year-over-year sales expansion.

Category performance: Technology leads sales at approximately \$836K and profit at approximately \$145K. Office Supplies follows in profit, while Furniture produces substantially lower profit relative to its sales volume.

Profitability warning: Some sub-categories are loss-making. Tables show the largest visible loss, making them a priority for pricing, cost, product-mix, or discount review.

Regional performance: The West region leads both sales and profit in the dashboard, followed by East.

Business principle: High sales do not automatically imply high profitability; decisions should consider both revenue and margin.

13. Recommendations

- Review pricing, cost, and discount strategies for low-margin and loss-making products.
- Investigate the drivers behind the weak profitability of Furniture and specific loss-making sub-categories.
- Protect successful high-profit categories while monitoring discount levels.
- Study the practices behind strong regional performance and evaluate whether they can be replicated elsewhere.
- Use the dashboard as a recurring management report rather than relying only on static monthly summaries.

14. Conclusion

This project demonstrates a complete analytical workflow: preparing transactional data, building a dimensional model, developing DAX measures, validating results, designing interactive dashboards, and translating findings into business recommendations. The final deliverable is structured for portfolio presentation and can be extended with additional customer-retention, forecasting, or profitability analyses.

Project Deliverables

- Power BI report (.pbix)
- Final dashboard screenshots
- Project documentation PDF
- GitHub repository and README
- Portfolio / CV project entry

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